SR UNIVERSITY

School of Computer Science and Artificial Intelligence

Lab Assignment # 3

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Question 1: Zero-Shot Prompting (Palindrome Number Program)
Write a zero-shot prompt (without providing any examples) to generate a Python function that checks whether a given number is a palindrome.
Task:
• Record the AI-generated code.
• Test the code with multiple inputs.
• Identify any logical errors or missing edge-case handling.

Code:-

```
lab3_2.py > ...
#Write a Python function to check whether a given number is a palindrome or not.
#The function should take an integer as input and return True if it is a palindrome, otherwise return False.
#Also handle special cases like negative numbers and numbers ending with zero.
#--Task 1--
from logging import root
def palindrome_number(n):
    if n < 0:
        return False
    str_n = str(n)
    return str_n == str_n[::-1]
print(palindrome_number(121))
print(palindrome_number(-121))
print(palindrome_number(10))
print(palindrome_number(12321))
```

Output:-

```
True
False
True
True
False
```

Question 2: One-Shot Prompting (Factorial Calculation)
Write a one-shot prompt by providing one input-output example and ask the AI to generate a Python function to compute the factorial of a given number.
Example:
Input: 5 → Output: 120
Task:
• Compare the generated code with a zero-shot solution.

• **Examine improvements in clarity and correctness.**

Prompt:-

input: 5 => output: 120 write a function to calculate factorial of a number

Code:-

```python
#-Task 2--
#input: 5 => output: 120 write a function to calculate factorial of a number
#code in input manner
def factorial(n):
    if n < 0:
        return "Factorial not defined for negative numbers"
    elif n == 0 or n == 1:
        return 1
    else:
        result = 1
        for i in range(2, n + 1):
            result *= i
        return result
try:
    num = int(input("Enter a number to calculate its factorial: "))
    print(f"The factorial of {num} is {factorial(num)}")
except ValueError:
    print("Invalid input. Please enter a valid integer.")
#--Task 3--
```

Output:-

```
Enter a number to calculate its factorial: 5
The factorial of 5 is 120
```

Question 3: Few-Shot Prompting (Armstrong Number Check)
Write a few-shot prompt by providing multiple input-output examples to guide the AI in generating a Python function to check whether a given number is an Armstrong number.
Examples:
- **Input: 153 → Output: Armstrong Number**
- **Input: 370 → Output: Armstrong Number**
- **Input: 123 → Output: Not an Armstrong Number**

Task:
- **Analyze how multiple examples influence code structure and accuracy.**
- **Test the function with boundary values and invalid inputs.**

(Optional Extension)

Prompt:-

input: 370 => output: Armstrong write a function to check if a number is an Armstrong number.

Code:-

```python
#--Task 3--
#input: 370 => output: Armstrong write a function to check if a number is an Armstrong number.
#in inpur manner
def is_armstrong(n):
    num_str = str(n)
    num_digits = len(num_str)
    sum_of_powers = sum(int(digit) ** num_digits for digit in num_str)
    return sum_of_powers == n
try:
    number = int(input("Enter a number to check if it is an Armstrong number: "))
    if is_armstrong(number):
        print(f"{number} is an Armstrong number.")
    else:
        print(f"{number} is not an Armstrong number.")
except ValueError:
    print("Invalid input. Please enter a valid integer.")
```

Output:-

```
Enter a number to check if it is an Armstrong number: 153
153 is an Armstrong number.
```

Question 4: Context-Managed Prompting (Optimized Number Classification)

Design a context-managed prompt with clear instructions and constraints to generate an optimized Python program that classifies a number as prime, composite, or neither.

Task:

- **Ensure proper input validation.**
- **Optimize the logic for efficiency.**
- **Compare the output with earlier prompting strategies.**

Prompt:-

write a program on a context-managed that classifies number as prime, composite or neither in input manner the python code.

Code:-

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```python
#write a program on a context-managed that classifies number as prime, composite or neither
#in input manner the python code.
def classify_number(n):
    if n <= 1:
        return "Neither prime nor composite"
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return "Composite"
    return "Prime"
try:
    num = int(input("Enter a number to classify: "))
    classification = classify_number(num)
    print(f"{num} is classified as: {classification}")
except ValueError:
    print("Invalid input. Please enter a valid integer.")
```

Output:-

```
Enter a number to classify: 10
10 is classified as: Composite
PS C:\sagar\pack1\OneDrive\3-2\AI Assistant Codi
nt Coding/lab3_2.py"
Enter a number to classify: 1
1 is classified as: Neither prime nor composite
```

Question 5: Zero-Shot Prompting (Perfect Number Check)
Write a zero-shot prompt (without providing any examples) to generate a Python function that checks whether a given number is a perfect number.
Task:
- **Record the AI-generated code.**
- **Test the program with multiple inputs.**

- **Identify any missing conditions or inefficiencies in the logic.**

Code:-

```python
#--Task 5--
#write a python program prefect number true or false
#in input manner
def is_perfect_number(n):
    if n <= 0:
        return False
    divisors_sum = sum(i for i in range(1, n) if n % i == 0)
    return divisors_sum == n
try:
    number = int(input("Enter a number to check if it is a perfect number: "))
    if is_perfect_number(number):
        print(f"{number} is a perfect number.")
    else:
        print(f"{number} is not a perfect number.")
except ValueError:
    print("Invalid input. Please enter a valid integer.")
```

Output:

```
Enter a number to check if it is a perfect number: 6
6 is a perfect number.
```

Question 6: Few-Shot Prompting (Even or Odd Classification with Validation)

Write a few-shot prompt by providing multiple input-output examples to guide the AI in generating a Python program that determines whether a given number is even or odd, including proper input validation.

Examples:

• Input: 8 → Output: Even
• Input: 15 → Output: Odd
• Input: 0 → Output: Even

Task:

• Analyze how examples improve input handling and output clarity.
• Test the program with negative numbers and non-integer inputs.

Prompt:-

input: 8 -> output: Even
input: 15 -> output: Odd
input: 0 -> output: Even
write a function to check if a number is even or odd.

Code:-

Assignment3.py > ...

```python
#Task6
# input: 8 -> output: Even
# input: 15 -> output: Odd
# input: 0 -> output: Even write a function to check if a number is even or odd.
def even_or_odd(number):
    return "Even" if number % 2 == 0 else "Odd"
print(even_or_odd(8))
print(even_or_odd(6.2))
print(even_or_odd(6/3))
print(even_or_odd(5**0.5))
print(even_or_odd(3.14159))
```

Output:-

```
Enter a number to check if it is even or odd: 8
8 is Even.
PS C:\sagar\pack1\OneDrive\3-2\AI Assistant Coding
nt Coding/lab3_2.py"
Enter a number to check if it is even or odd: 5
5 is Odd.
```